**9** The times that 48 trains left a station on Monday were recorded.

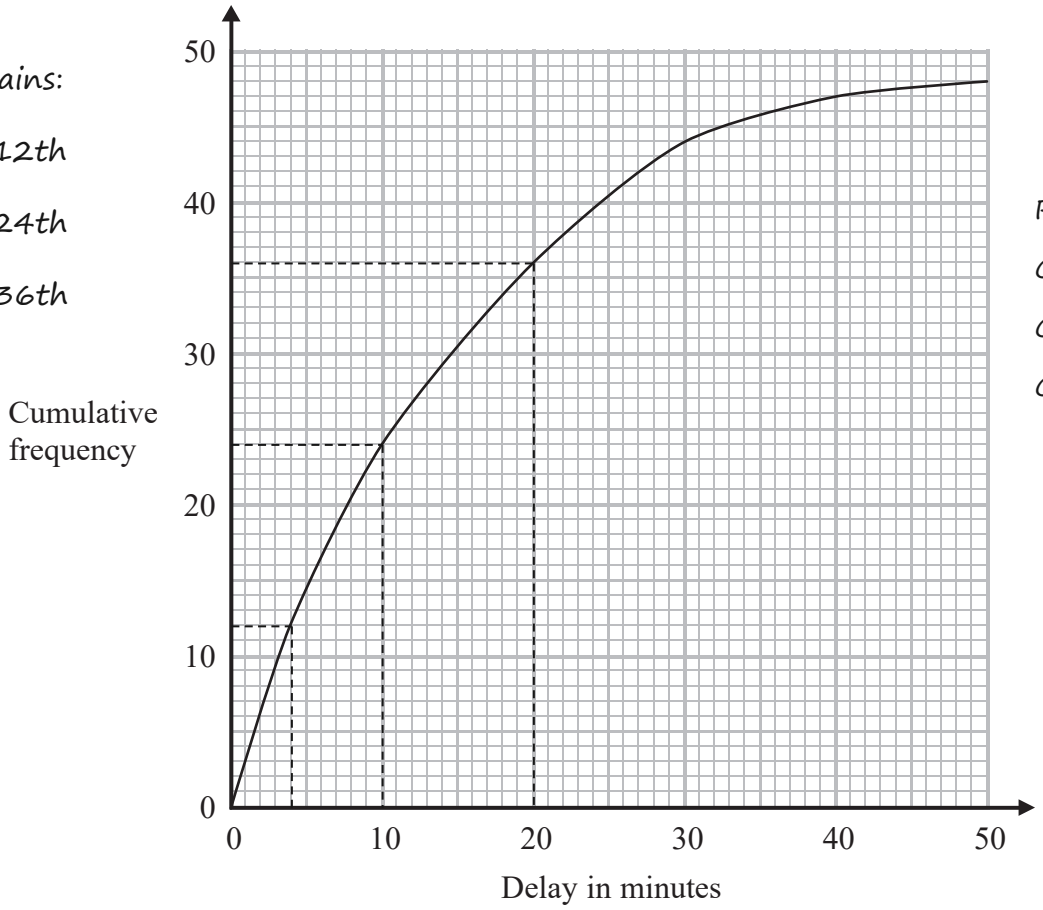
The cumulative frequency graph gives information about the numbers of minutes the trains were delayed, correct to the nearest minute.

48 trains:

¼ → 12th

½ → 24th

¾ → 36th




Read:

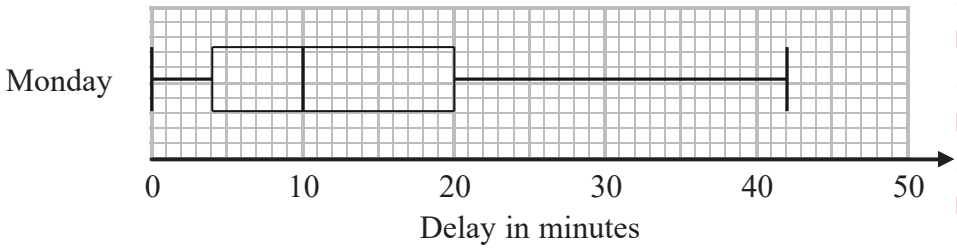
$Q_1 = 4$

$Q_2 = 10$

$Q_3 = 20$

The shortest delay was 0 minutes.
The longest delay was 42 minutes.

(a) On the grid below, draw a box plot for the information about the delays on Monday.



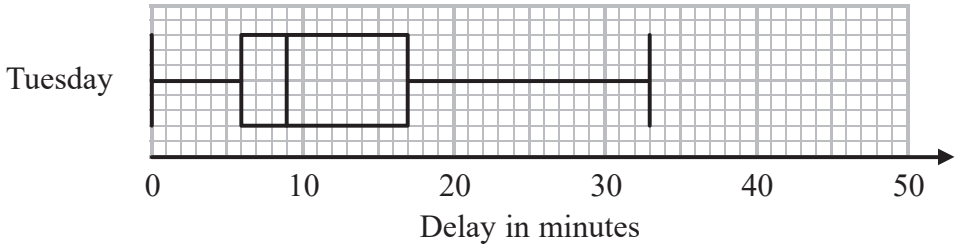

**1 mark**
Ends: 0, 42

**1 mark**
Median: 10

**1 mark**
Box: 4 to 20

**(3)**

48 trains left the station on Tuesday.
The box plot below gives information about the delays on Tuesday.




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(b) Compare the distribution of the delays on Monday with the distribution of the delays on Tuesday.

*Monday has the greater median delay: 10 min versus 9 min.*

**1 mark**

Compare typical delays in context

*Monday also has the greater range: 42 min versus 33 min.*

**1 mark**

Monday's delays are more spread out

**(2)**

Mary says,

"The longest delay on Tuesday was 33 minutes.
This means that there must be some delays of between 25 minutes and 30 minutes."

(c) Is Mary right?
You must give a reason for your answer.

*No. The longest 12 delays could all be between 17 and 25*

*or between 30 and 33 minutes, with none between 25 and 30.*

**1 mark**

A box plot does not locate every individual delay

**(1)**

**(Total for Question 9 is 6 marks)**

**10** (a) Simplify $\dfrac{x-1}{5(x-1)^2}$

**(1)**

(b) Factorise fully $50 - 2y^2$

**(2)**

**(Total for Question 10 is 3 marks)**

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